Auteur: Seda Jusjajeva
Studierichting: Informatica
Oefeningenreeks 1B nr. 43.2

$z^2 - 2z + 1 - 2i = 0$

$D = 4 - 4 \cdot (1 - 2i) = 4 - 4 + 8i = 8i$

$\Leftrightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = 8 \end{cases}$

$\Rightarrow \begin{cases} x^2 - y^2 = 0 \\ x^2(-y^2) = -16 \end{cases}$

$\Rightarrow \lambda^2 - 16 = 0$

$\Leftrightarrow \lambda = \pm 4$

$\Leftrightarrow x = \pm 2$ en $y = \pm 2$

$z_1 = \dfrac{2 + (2 + 2i)}{2} = 2 + i$

$z_2 = \dfrac{2 - (2 - 2i)}{2} = -i$

References